

Sulfur isotopic features of sulfide ores in mafic intrusions of Norilsk region

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Basics about sulfur isotopy

Isotope	Terrestrial abundance* in %
^{32}S	95.02
^{33}S	0.75
^{34}S	4.21
^{36}S	0.02

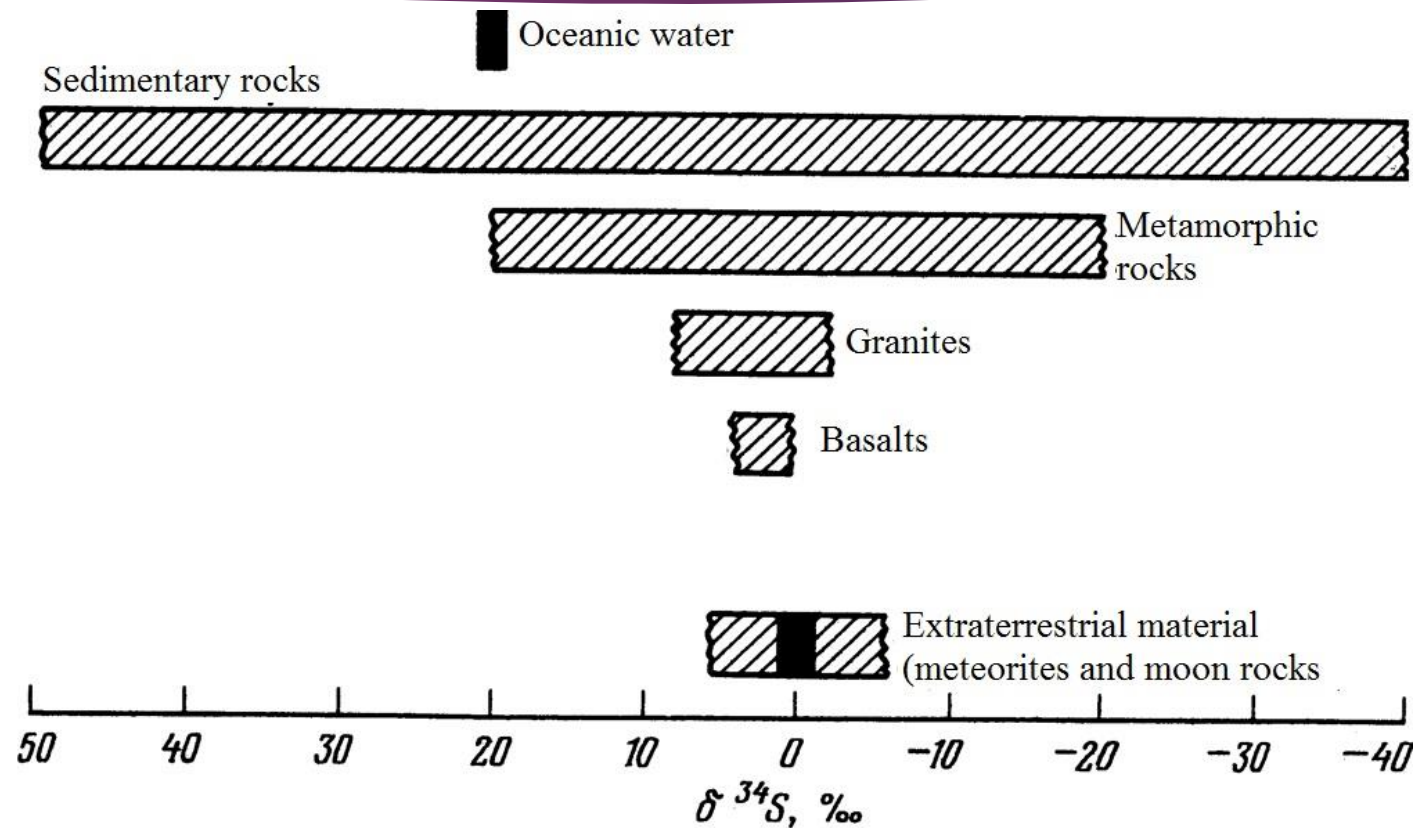
The δ -notation for the $^{34}\text{S}/^{32}\text{S}$

$$\delta^{34}\text{S} = \left(\frac{(^{34}\text{S}/^{32}\text{S})_{\text{sample}} - (^{34}\text{S}/^{32}\text{S})_{\text{reference}}}{(^{34}\text{S}/^{32}\text{S})_{\text{reference}}} \right) \times 1000$$

expressed in ‰ (“per mil”)

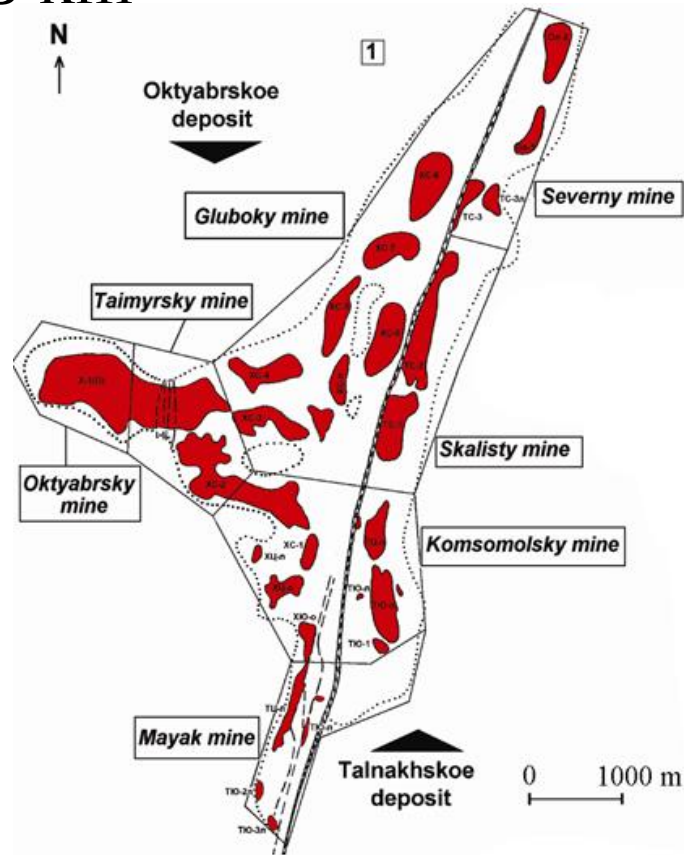
“+” (positive) values – more ^{34}S
“-” (negative) values – more ^{32}S
 ≈ 0 – closeness to reference standard

Values of $\delta^{34}\text{S}$ in different rocks

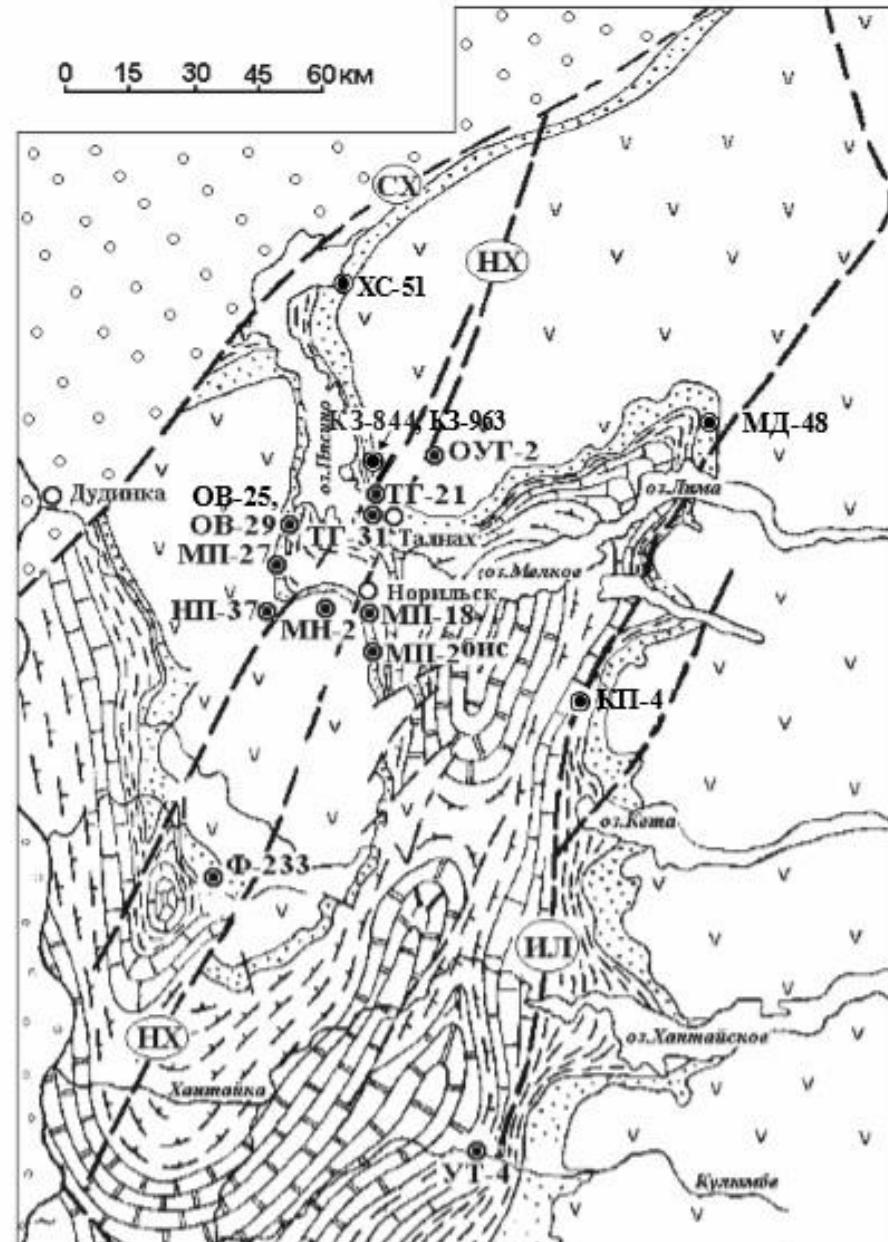


Norilsk copper-nickel deposits are unique by content of nickel, copper, cobalt and PGE. But the most important thing that all this reserves are concentrated in the territory of 8x18 km

Deposits associated with differentiated mafic-ultramafic intrusions



Position of drilling wells



Exploration criteria

- Structural
- Magmatic
- Mineralogical
- Geochemical
- Lithological
- Metamorphical
- Hydrogeochemical
- Geophysical
- Geomorphological
- Isotopic-geochemical *New*

Beginning of sulfur isotopy study

- ▶ In 1985 by Grinenko L.N. was described different sulfur isotopy features for different intrusions of Norilsk area.
- ▶ Naldrett A.J., Lightfoot P., Doherty W. and other foreign geologists studied Norilsk ores
- ▶ Most well-known Russian geologists studied Norilsk – Distler V.V., Rad'ko V.A., Sharkov E.V., Malitch K.N. et al.

$\delta^{34}\text{S}$ in sulfide ores of Norilsk intrusions

Intrusion	Number	Average	Min	Max	Standard deviation
Norilsk-1 (Dissem.)	16	9,41	5,5	16,7	2,56
Talnakh (Dissem.)	49	10,92	8,7	15	0,89
Talnakh (Massive)	49	10,99	7,8	14	0,92
Kharaelakh (Dissem.)	18	12,13	6,3	13,5	1,6
Kharaelakh (Massive)	35	12,25	8,4	13,6	1,06
Chernogorsky (Dissem.)	4	10,88	10,4	11,2	0,34
Zubovsky (Dissem.)	11	1,85	0	5,7	2,18
Vologochansky (Dissem.)	13	6,80	4	8,5	1,43
Uzhnopyasinsky (Dissem.)	7	8,36	4,3	10,5	2
Imangdinsky (Dissem.)	6	10,5	6,4	18,8	5,28
Mikchangdinsky (Dissem.)	4	12,78	11	14	1,28
Binudinsky (Dissem.)	7	2,14	0,7	4,7	1,33
Dumtaleisky (Dissem.)	15	11,37	9,9	12,9	0,76
Nizhnetalnakhsky (Dissem.)	8	7,16	6,5	8	0,5
Nizhnenorilsky (Dissem.)	15	4,96	3,8	7,7	0,97

1. Rich

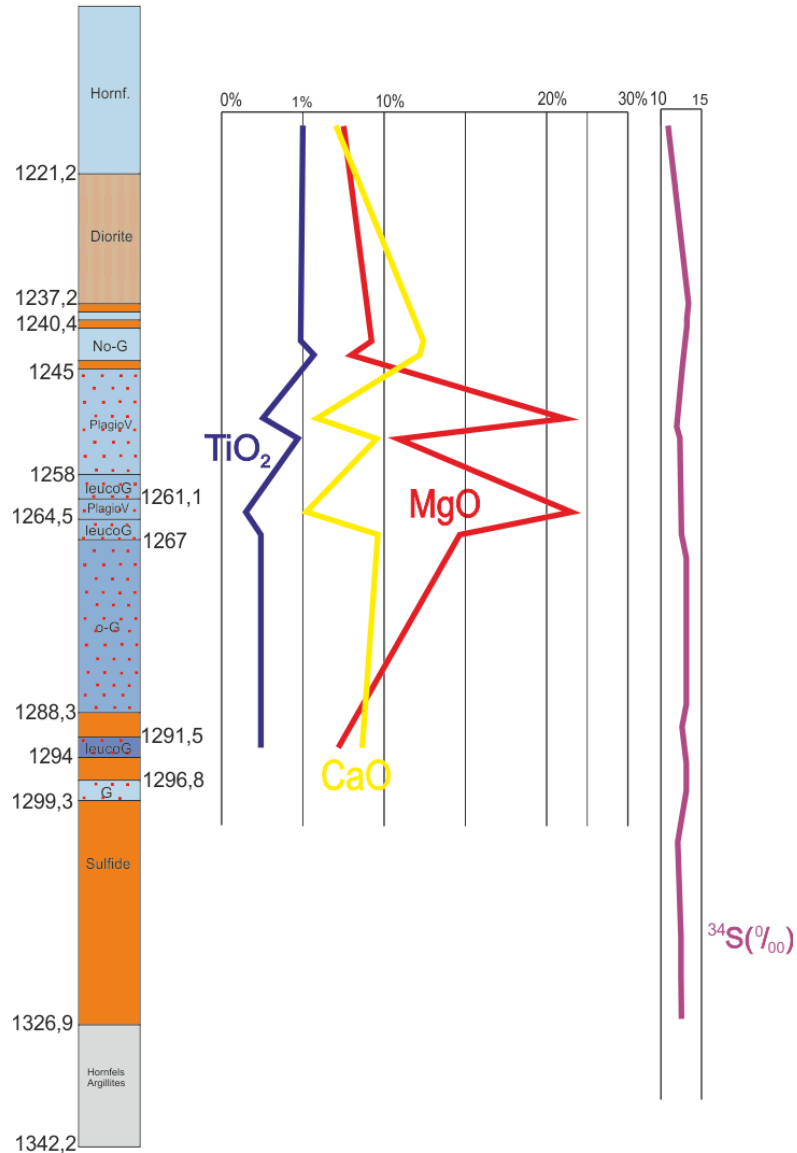
2. Productive

3. Potential

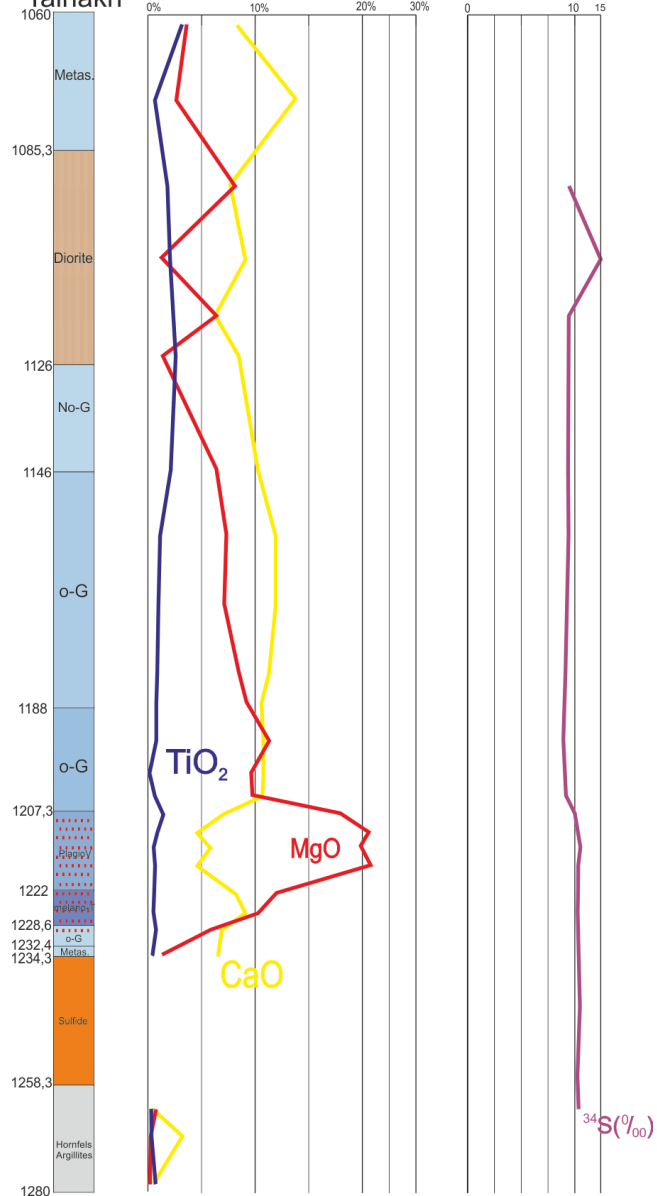
4. Underproductive

Value $\delta^{34}\text{S}$ don't change critically by depth

KZ-963
Kharaelakh



OYG-2
Talnakh



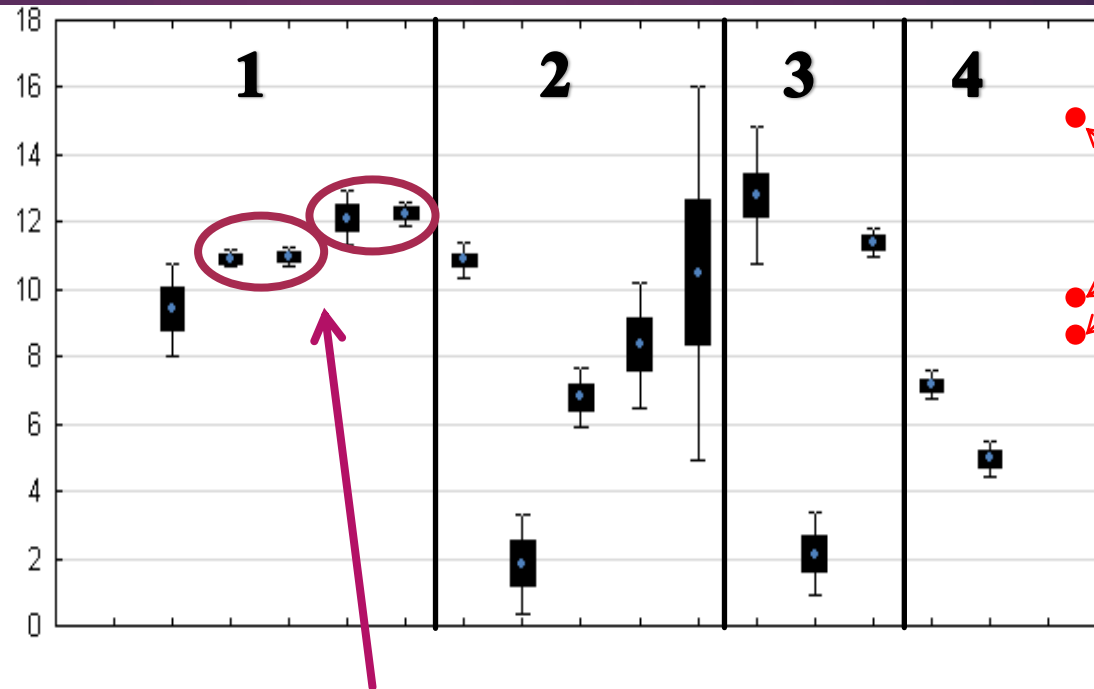
$\delta^{34}\text{S}$ in sulfide ores of Norilsk intrusions

1. Rich

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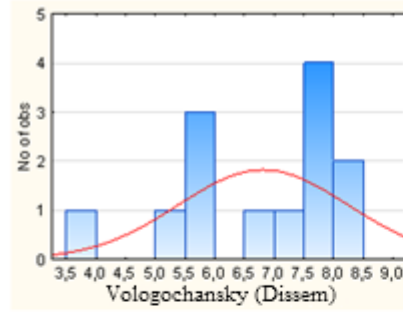
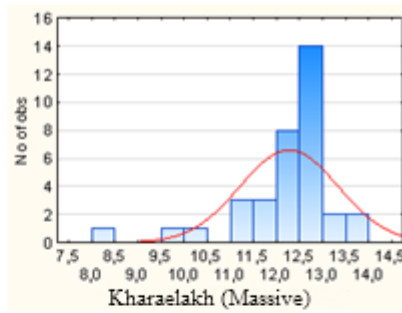
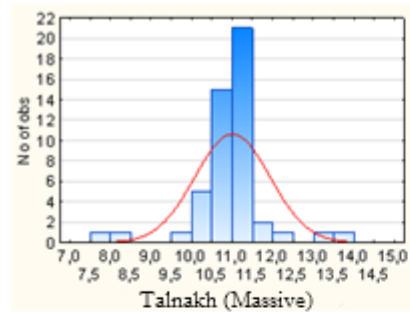
Values $\delta^{34}\text{S}$ of massive and disseminated ores in one intrusion are the same
Evidence of end of sulfur isotopes fractionation on the stage of magmatic melt

Several samples of low-grade productive and under-productive intrusions shows high values $\delta^{34}\text{S}$

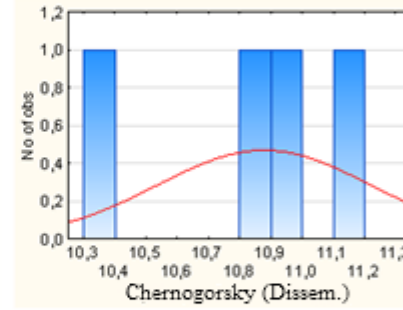
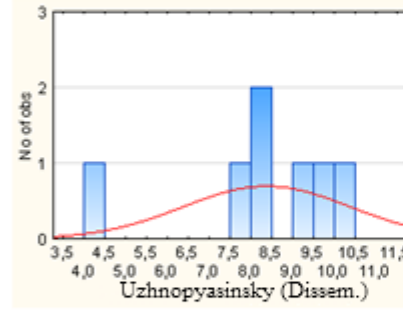
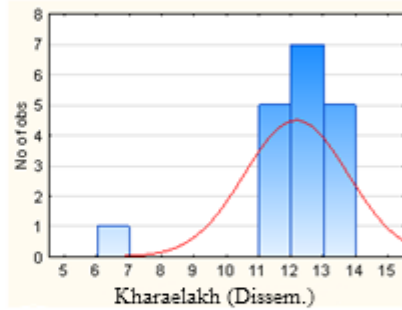
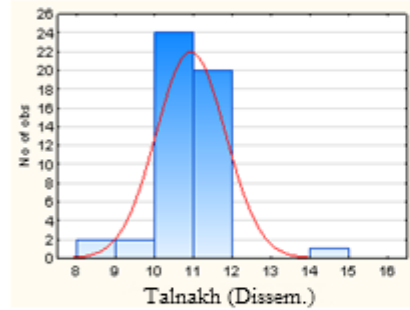
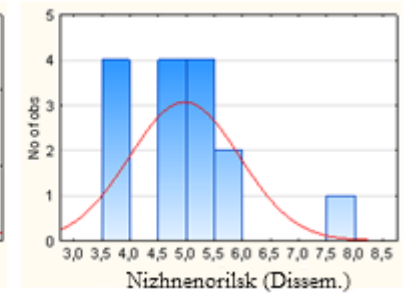
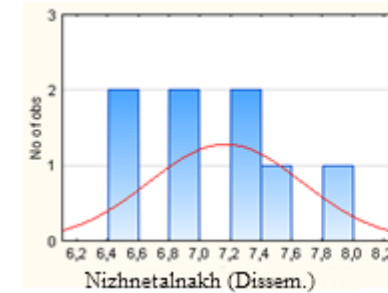
Which may indicate:

- Magma contamination with crust material on crust-mantle level (with end of fractionation?)
- High values of $\delta^{34}\text{S}$ the result of oxidation-reduction processes or fractionation

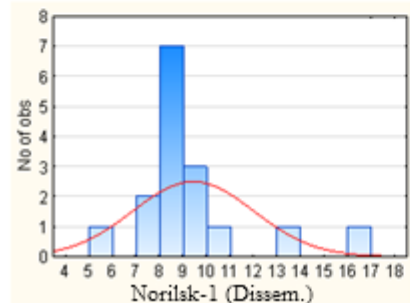
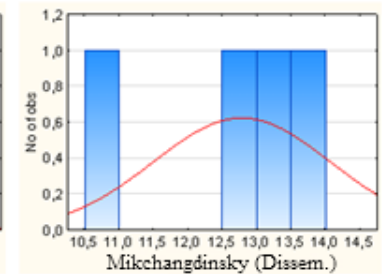
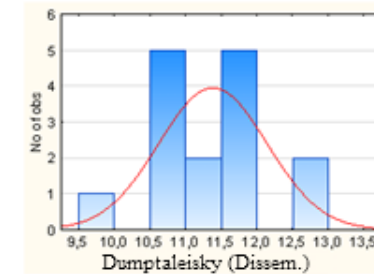
Homogeneous composition $\delta^{34}\text{S}$ and correlation with mineralization scale



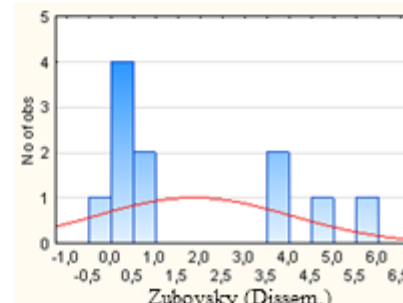
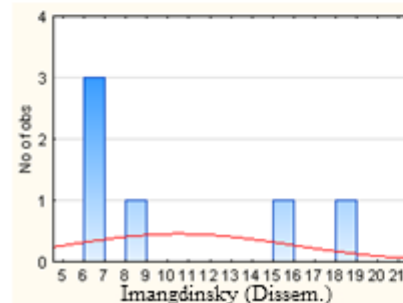
Productive



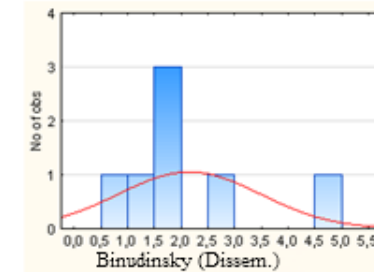
Under-productive



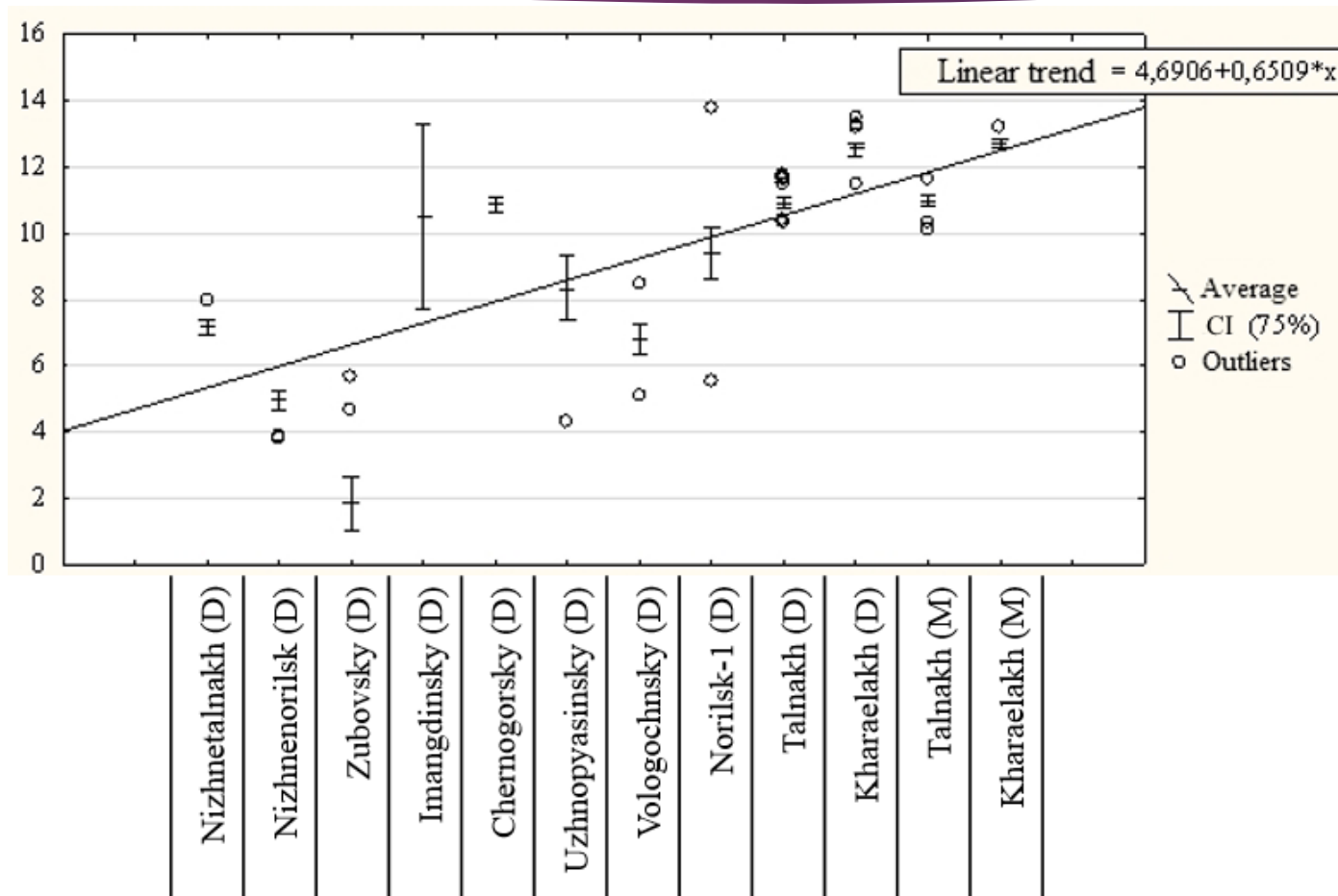
Rich



Potential



Homogeneous composition $\delta^{34}\text{S}$ and correlation with mineralization scale



Conclusions

1. Mean values of the isotopic composition of sulfur for rich intrusions in all types of sulfide copper-nickel ores (massive, disseminated and low-sulfide) are close.
2. Most of the ore-bearing intrusions are characterized by a high content of heavy isotope of sulfur ($\delta^{34}\text{S}$ - 11-12 ‰).
3. For under-productive intrusions characteristic of lower values of the heavy isotope of sulfur ($\delta^{34}\text{S}$ - 5-7 ‰), although here is exceptions.
4. Potentially ore-bearing intrusive have wide range of $\delta^{34}\text{S}$, there are both high and low values.
5. Given all the features, $\delta^{34}\text{S}$ cannot be used as direct exploration criteria.
6. The isotopic composition of sulfur may be a feature of the intrusive and may not be associated with an enrichment of magma by heavy isotope of sulfur from the earth's crust.
7. Statistical studies clearly indicate the homogeneity of the isotopic characteristics of the sulfur in the ore-mining industrial intrusions in comparison with others.
8. Reveals a trend line depending of the heavy isotope of sulfur content and scale mineralization.

Thank you for attention

